

Online Supplement to Characteristic-Space Metrics in Factor Models: Identification and Inference

Mingyang Liu

Imperial Business School, Imperial College London

September 2026

The main paper contains the assumptions and proofs needed for its results. This optional supplement gives matrix and profile derivative identities, additional reporting examples, computational checks, and the simulation designs and additional results. The separate replication package provides source code, frozen inputs, archived records, and validation logs.

A Matrix Derivatives and Chart Checks

This appendix supplies implementation identities for the explicit local chart in the main paper. The main results do not require a separate geometric construction from this supplement.

A.1 Metric and normalization derivatives

Let $Q \succ 0$, $S = Q^{1/2}$, and ΔQ be symmetric. Differentiating $S^2 = Q$ shows that $\dot{S} = D(Q^{1/2})[\Delta Q]$ solves the Sylvester equation

$$S\dot{S} + \dot{S}S = \Delta Q. \quad (\text{A.1})$$

If $S = V \text{diag}(s_1, \dots, s_L)V'$ with $s_i > 0$, then $(V'\dot{S}V)_{ij} = (V'\Delta QV)_{ij}/(s_i + s_j)$. Positive denominators make this derivative well defined even at repeated positive eigenvalues. The inverse derivative is $D(Q^{-1/2})[\Delta Q] = -S^{-1}\dot{S}S^{-1}$. These are matrix-function derivatives, not derivatives of a selected eigenvector.

Trace normalization $g(Q) = LQ/\text{tr}(Q)$ has derivative

$$Dg_Q[\Delta Q] = \frac{L}{\text{tr}(Q)} \left\{ \Delta Q - \frac{\text{tr}(\Delta Q)}{\text{tr}(Q)} Q \right\}. \quad (\text{A.2})$$

Applying it to each centered Gram contribution gives the influence of the normalized metric. The normalization must be included even when a particular report is scale invariant.

A.2 The graph chart

Use the main paper's reference matrices $U_0, U_{0\perp}$ and write $U(C) = (U_0 + U_{0\perp}C)(I + C'C)^{-1/2}$. Then $U(C)'U(C) = I_K$, since $(U_0 + U_{0\perp}C)'(U_0 + U_{0\perp}C) = I + C'C$. At $C = 0$ the derivative of $C'C$ is zero, giving

$$DU(0)[\dot{C}] = U_{0\perp}\dot{C}, \quad D_C B(0; Q)[\dot{C}] = Q^{-1/2}U_{0\perp}\dot{C}. \quad (\text{A.3})$$

The retained-block map projects onto the Q -orthogonal complement:

$$A(C, D; Q) = (I - BB'Q)(A_0 + Q^{-1/2}U_{0\perp}D).$$

Consequently $B'QA = 0$. At the reference metric and origin, $B = B_0$ and $A = A_0$. The inverse formulas in the main paper show that this map is locally one-to-one on loading classes, not on all rotated bases individually.

A fixed orthogonal change of the reference complement changes (C, D) but leaves the represented classes unchanged. A metric-dependent change of chart also changes its metric derivative. The report chain rule must then include that transition derivative; dropping it would compare different partial derivatives rather than the same report.

A.3 Numerical validation

Useful checks compare analytic derivatives with central differences at several step sizes, verify $B'QB = I$ and $B'QA = 0$, and reconstruct the same class after changing the reference basis. An explicit local chart is a differentiation device, not an estimator by itself. The companion code separately tests profile derivatives, quotient curvature, complete influences, and the polished score condition.

B Additional Profile Calculations

B.1 Least squares in supported feature space

For any candidate index h , let $X = N^{-1}Z'r$, $W = N^{-1}Z'Z$, and $m = X - Wh$. Since $\text{Range}(W) = \text{Range}(Z')$,

$$\begin{aligned}\frac{1}{2}m'W^+m &= \frac{1}{2}X'W^+X - X'h + \frac{1}{2}h'Wh, \\ \frac{1}{2N}\|r - Zh\|^2 &= \frac{1}{2N}r'r - X'h + \frac{1}{2}h'Wh.\end{aligned}\tag{B.1}$$

The two criteria therefore differ only by a data-dependent constant. A zero eigenvalue of W contributes to neither the managed moment nor the fitted return. This is an exact objective equivalence, independent of a noise-efficiency interpretation.

B.2 From the envelope score to the Hessian

Fix a date and omit its index temporarily. At the exact profile define $f^* = f^*(B, A)$, $\tilde{m} = X - W(Bf^* + Ac)$, and $G = (B'WB)^{-1}$. For a loading perturbation let

$$\begin{aligned}\dot{f} &= G\{\dot{B}'\tilde{m} - B'W(\dot{B}f^* + \dot{A}c)\}, \\ \dot{h} &= \dot{B}f^* + \dot{A}c + B\dot{f}, \quad \dot{\tilde{m}} = -W\dot{h}.\end{aligned}\tag{B.2}$$

The matrix gradients are $\nabla_B \tilde{q} = -\tilde{m}f^{*'}$ and $\nabla_A \tilde{q} = -\tilde{m}c'$. The product rule gives

$$D\nabla_B \tilde{q} = W\dot{h}f^{*'} - \tilde{m}\dot{f}', \quad D\nabla_A \tilde{q} = W\dot{h}c'.\tag{B.3}$$

The term involving $\dot{f}$ is essential even though the first derivative used the envelope theorem.

For stacked ambient coordinates ϑ and chart Jacobian $J_\eta = D_\eta \vartheta(\eta; Q)$, the quotient Hessian has bilinear form

$$D_\eta^2 \tilde{Q}_T[v, w] = (J_\eta v)' D_\vartheta^2 \tilde{\mathcal{L}}_T(J_\eta w) + \nabla_\vartheta \tilde{\mathcal{L}}_T D_\eta^2 \vartheta[v, w].\tag{B.4}$$

The second term is chart curvature. It cannot be deleted merely because the quotient score vanishes; its deletion requires the corresponding ambient stationarity argument. Retaining the full expression is valid at generic sample and population arguments.

B.3 Structural score centering

At structural loadings, set $G^\circ = (B^{\circ'}WB^\circ)^{-1}$ and $R^\circ = I - WB^\circ G^\circ B^{\circ'}$. The identities $R^\circ WB^\circ = 0$ and $(R^\circ)^2 = R^\circ$ follow by direct multiplication. Generally $R^{\circ'} \neq R^\circ$: the residual projector is oblique in Euclidean coordinates.

The profile gives $f^* = f^\circ + G^\circ B^{\circ'}u$ and $\tilde{m} = R^\circ u$. Hence the loading score is

$$\begin{aligned}D\tilde{q}[\dot{B}, \dot{A}] &= - (R^\circ u)' (\dot{B}f^\circ + \dot{A}c) \\ &\quad - (R^\circ u)' \dot{B}G^\circ B^{\circ'}u.\end{aligned}\tag{B.5}$$

Conditional mean zero eliminates the first line. If $\mathbb{E}(uu' \mid \mathcal{C}) = \omega W$, the conditional

expectation of the second line equals

$$\begin{aligned} -\omega \operatorname{tr}(R^{\circ'} \dot{B} G^{\circ} B^{\circ'} W) &= -\omega \operatorname{tr}(G^{\circ} B^{\circ'} W R^{\circ'} \dot{B}) \\ &= 0, \end{aligned}$$

since $B^{\circ'} W R^{\circ'} = (R^{\circ} W B^{\circ})' = 0$. The main paper explains how integrability, deterministic date weights, and the chart chain rule convert this into population score centering. Strict slackness and positive quotient curvature then yield local structural-target coincidence.

B.4 The strength of profile regularity

A lower bound uniform over all sample dates is not implied by positive definiteness at each date. With independent Gaussian cross sections of fixed size, the smallest supported eigenvalue has support arbitrarily close to zero. For a fixed positive threshold with positive lower-tail probability, the probability that all T dates exceed it decreases rather than tends to one. This explains the explicit qualification of the Gaussian simulation in the main paper. Replacing the bound by inverse moments would require proving the needed derivative envelopes and uniform laws under that alternative condition.

C Additional Reports and Covariance Details

C.1 Reports beyond a single alignment

A report is a prespecified smooth map of the loading class and declared metric. For a fixed vector a in whitened coordinates, $a' \mathcal{P}_Q(B) a$ differs from the main paper's characteristic-direction alignment, which first whitens and normalizes a fixed raw direction. Both are valid maps, but they hold different benchmarks fixed as Q changes. The profiled-estimator simulation uses the former.

A fixed full-rank matrix $D_{\star}$ gives a benchmark projector. The squared distance between projectors compares entire spans without selecting eigenvectors or ordering latent factors. A fixed-weight vector report can collect alignment, span distance, and intercept exposure; a contrast $r_{\phi} = (r_1, r_2, r_3)'$ specifies in advance which linear combination is being estimated. The joint covariance retains the dependence between these three questions because they use the same estimated loading class and metric.

C.2 Block attribution and overlap

On a neighborhood where the observed loading block Γ_δ has fixed full column rank, define

$$P_{\beta,Q} = \mathcal{P}_Q(B), \quad P_{\delta,Q} = \mathcal{P}_Q(\Gamma_\delta), \quad u_{\alpha,Q} = Q^{1/2}\Gamma_\alpha.$$

A descriptive block report may include $\text{vec}(P_{\beta,Q})$, $\text{vec}(P_{\delta,Q})$, $\|u_{\alpha,Q}\|^2$, and $u'_{\alpha,Q}P_{\delta,Q}u_{\alpha,Q}$. The last quantity measures intercept exposure aligned with the observed loading span.

Under the maintained equality restrictions, however,

$$\text{tr}(P_{\beta,Q}P_{\delta,Q}) = 0, \quad u'_{\alpha,Q}P_{\beta,Q}u_{\alpha,Q} = 0. \quad (\text{C.1})$$

These are imposed identities, not informative statistical tests of separate identification. Nor do pairwise overlap measures form an additive variance decomposition. Local identification requires the population score and curvature conditions in the main paper. This distinction prevents a geometric accounting identity from being reported as evidence of independent economic risks.

C.3 What a zero first derivative means

Smoothness permits a first-order expansion but does not ensure a nondegenerate first-order law. If a squared projector distance is evaluated at identical spans, its derivative is zero. Likewise, at $\Gamma_{\alpha,0} = 0$,

$$D(\Gamma'_\alpha Q \Gamma_\alpha)[\dot{\Gamma}_\alpha, \Delta Q] = 0.$$

The loading estimator can still be regular; the report then starts at second order. At a positive slack budget its exposure limit is the quadratic form stated in the main paper. At zero budget, the parameter space itself is lower dimensional and needs a different analysis.

C.4 Influence replacement for HAC

For true influences ζ_t and estimation errors $e_t = \widehat{\zeta}_t - \zeta_t$, a lag-product difference is

$$\widehat{\zeta}_t \widehat{\zeta}'_{t-\ell} - \zeta_t \zeta'_{t-\ell} = e_t \zeta'_{t-\ell} + \zeta_t e'_{t-\ell} + e_t e'_{t-\ell}.$$

Cauchy–Schwarz bounds the norm of its average, uniformly over the lag, by $2\rho_T Z_T + \rho_T^2$, where $\rho_T^2 = T^{-1} \sum_t \|e_t\|^2$ and $Z_T^2 = T^{-1} \sum_t \|\zeta_t\|^2 = O_p(1)$. Summing bounded kernel weights over b_T lags gives an $O_p(b_T \rho_T + b_T \rho_T^2)$ replacement bound. Thus derivative consistency without a rate does not, by itself, justify a diverging-bandwidth plug-in HAC estimator.

The same construction applies to the stacked score and metric influences used in boundary simulation. The covariance must be estimated jointly; estimating only its marginal blocks would delete the dependence that affects the projected loading movement and random boundary shift.

D Computing the Learned-Boundary Law

The main paper states and proves the boundary approximation and uniform coverage theorem. This appendix records useful implementation identities.

D.1 Exposure derivatives

Let $\Gamma_\alpha(\eta, Q)$ be the intercept representative. At the unrestricted reference target, write $\dot{\Gamma}_{\alpha,\eta} = D_\eta \Gamma_\alpha[v]$ and $\dot{\Gamma}_{\alpha,Q} = D_Q \Gamma_\alpha[\Delta Q]$. The boundary derivatives are

$$\begin{aligned} \gamma'_\eta v &= 2\Gamma'_\alpha Q \dot{\Gamma}_{\alpha,\eta}, \\ \gamma'_Q \text{vec}(\Delta Q) &= 2\Gamma'_\alpha Q \dot{\Gamma}_{\alpha,Q} + \Gamma'_\alpha \Delta Q \Gamma_\alpha. \end{aligned} \tag{D.1}$$

Holding the representative fixed removes the first term in the second line and generally changes the boundary being approximated. The exposure budget is deterministic in this experiment; estimating it would add another influence contribution.

D.2 Projection and endpoint computation

For each joint Gaussian draw define

$$\begin{aligned} v'' &= -H^{-1} \mathcal{Z}, & A &= \gamma'_\eta v'' + \gamma'_Q \mathcal{Z}_Q, \\ U &= \phi_\eta v'' + \phi_Q \mathcal{Z}_Q, & c &= \frac{\phi_\eta H^{-1} \gamma_\eta}{\gamma'_\eta H^{-1} \gamma_\eta}. \end{aligned}$$

The scalar report is $L(\kappa) = U - c[A + \kappa]_+$. The interior endpoint is U and the binding endpoint is $U - c[A]_+$. Common draws therefore give a direct algorithm:

1. Estimate the full joint covariance and the profile, constraint, and report derivatives in one chart.
2. Draw one common sample of joint Gaussian disturbances.
3. Compute the two endpoint report samples and their critical quantiles.
4. Take the smaller lower-tail and larger upper-tail quantile, then invert the root- T report error to form the interval.

The monotonicity proof in the main paper establishes exact endpoint sufficiency for this scalar one-constraint map. Finite-grid refinement is unnecessary here. It remains useful as an independent implementation check: grid and endpoint envelopes must agree up to floating-point error.

D.3 Singular inputs and computational safeguards

If $\widehat{J} = V \text{diag}(d_j)V' \succeq 0$, simulate $V \text{diag}(\sqrt{d_j})e$ with independent standard-normal entries of e . Zero eigenvalues are valid. Clipping negative numerical eigenvalues to zero does not require inventing variation in redundant directions. Any positive eigenvalue floor must have negligible asymptotic effect.

Check that the projected draw satisfies its linearized inequality and that the KKT multiplier is nonnegative. With an inactive constraint the draw should equal the unrestricted movement. Under an invertible coordinate change, consistently transformed derivatives and Gaussian inputs should produce the same scalar report. These are algebraic and numerical checks, not substitutes for uniform localization or Gaussian approximation of the selected estimator.

A conventional bootstrap of a boundary estimator is not validated by these checks. Boundary bootstrap consistency requires reproducing the relevant local experiment, including the nuisance drift (Andrews, 2000; Fang and Santos, 2019).

E Profiled-Estimator Simulation Details

E.1 Design and interpretation

The baseline uses $L = 12$ characteristics, $K = 2$ latent factors, $K_o = 1$ observed factor, and $N = 100$ stocks per date. Let V be a fixed orthogonal matrix and let λ_j be twelve geometrically spaced values from 0.30 to 2.40. Trace-normalize $Q_0 = V \text{diag}(\lambda_j)V'$ to trace twelve. The condition number is eight. A second fixed orthogonal matrix $U = (u_1, \dots, u_{12})$ determines

$$B^\circ = Q_0^{-1/2}[u_1, u_2], \quad \Gamma_\alpha^\circ = 0.16Q_0^{-1/2}u_3, \quad \Gamma_\delta^\circ = Q_0^{-1/2}u_4. \quad (\text{E.1})$$

These formulas impose the declared normalization and cross-block orthogonality exactly. Orthogonal rotations are coordinate choices in this Gaussian construction; the spectrum, factor variances, and relative block magnitudes are substantive design choices, not without-loss normalizations. The intercept exposure is $0.16^2 = 0.0256$, well below the budget of five.

The latent factors have stationary variances $v_1 = 1$, $v_2 = 0.35$:

$$f_{j,t+1} = 0.6f_{j,t} + \sqrt{(1 - 0.6^2)v_j} e_{j,t+1}.$$

Independent standard-normal innovations and stationary initialization preserve these variances. The observed factor is $g_{t+1} = 0.6f_{1,t+1} + \sqrt{1 - 0.6^2} e_{g,t+1}$. Conditional on factors, the stock characteristics are independent $\mathcal{N}(0, Q_0)$ draws and disturbances are independent $\mathcal{N}(0, 0.6^2)$. These specifications generate overlap between observed and latent factor realizations without violating loading orthogonality in characteristic space.

The population design uses seed 2026080401. Coverage panels use NumPy SeedSequence with components (20251124, 31, T , replication) and separate child streams for factors, observed-factor noise, characteristics, and return disturbances. Replication identifiers 10000–11999 are used at each horizon. No failed draw is replaced by another estimator or a relaxed tolerance.

E.2 Frozen reports and the estimator

Candidate unit vectors are $a = (u_k + u_j)/\sqrt{2}$ for $k \in \{1, 2\}$ and $j \in \{5, \dots, 12\}$. Their projector reports all have population value 0.5. A separate 20,000-date calibration freezes

the candidate influences: the median nonmaximum calibrated variance selects the moderate target, and the maximum selects the high target. In one-based indices these are $(u_1 + u_8)/\sqrt{2}$ and $(u_2 + u_8)/\sqrt{2}$. These labels refer to the frozen candidate set, not an optimal reporting rule across all possible economic questions.

Each panel uses four alternating updates followed by zero-ridge quotient polishing of the exact profile. With quotient dimension $d = 40$, the stopping condition is $\|S_T\| \leq 10^{-8}\sqrt{d}$, with at most twenty polishing iterations. Thus 10^{-8} is a root-mean-square coordinate-score tolerance, not a bound on the unscaled Euclidean score norm. The recorded gates include the score norm, equality constraints, positive quotient curvature, and adjoint residual. The companion code also evaluates final intercept exposure explicitly. The all-date population profile-eigenvalue condition remains stronger than these finite-sample checks, as explained in the main paper.

Bartlett HAC uses bandwidth $\lfloor 1.5T^{1/3} \rfloor$: nine for $T = 240$ and fourteen for $T = 960$. The oracle-influence benchmark evaluates scores at the structural loadings and uses frozen reference derivatives from the separate long calibration; it is a numerical population-reference calculation, not a closed-form exact Hessian. The feasible method estimates these objects from the current panel.

E.3 Additional results

Figure S.1 plots the nominal 95 percent coverage comparison. The no-cross calculation retains marginal return and metric variance channels but removes their signed cross-covariance. In this design it lies close to complete-score HAC; that empirical feature does not impose orthogonality in the theory.

The studentized quantiles put numbers on the main paper’s quantile plots. The variance table answers a different question: whether the return and metric channels reinforce or offset one another. Cross terms are small in this design, explaining the similarity between the no-cross and complete coverage curves. Their small empirical size is not an imposed orthogonality condition. Together, these diagnostics distinguish scale correction from remaining departures from normality and finite-sample HAC error.

Table S.I: Additional profiled-estimator results, $T = 240$

Method	Mean SE	SE/SD	90% cov.	95% cov.
Moderate-sensitivity report				
MC scale	0.0058	1.000	90.35	94.90
Oracle HAC	0.0057	0.985	88.55	94.15
Fixed metric	0.0048	0.820	79.95	88.20
No cross term	0.0057	0.978	88.80	93.95
Complete HAC	0.0057	0.976	88.00	93.75
High-sensitivity report				
MC scale	0.0078	1.000	90.50	95.25
Oracle HAC	0.0075	0.957	88.05	93.60
Fixed metric	0.0065	0.829	82.15	88.15
No cross term	0.0074	0.948	87.35	93.40
Complete HAC	0.0073	0.942	86.95	93.20

Notes: Each entry summarizes 2,000 replications. SE/SD is the scale ratio; coverage is in percent, measured around the common population value 0.5. The Oracle HAC row uses frozen calibration derivatives at structural loadings.

Table S.II: Additional profiled-estimator results, $T = 960$

Method	Mean SE	SE/SD	90% cov.	95% cov.
Moderate-sensitivity report				
MC scale	0.0029	1.000	90.15	94.75
Oracle HAC	0.0029	1.004	89.75	94.90
Fixed metric	0.0024	0.838	83.35	89.05
No cross term	0.0029	0.997	89.40	94.65
Complete HAC	0.0029	0.996	89.25	94.25
High-sensitivity report				
MC scale	0.0037	1.000	90.10	95.10
Oracle HAC	0.0038	1.037	91.15	95.20
Fixed metric	0.0033	0.891	85.60	91.75
No cross term	0.0037	1.017	90.60	95.05
Complete HAC	0.0037	1.016	90.55	94.85

Notes: Each entry summarizes 2,000 replications. SE/SD is the scale ratio; coverage is in percent, measured around the common population value 0.5. The Oracle HAC row uses frozen calibration derivatives at structural loadings.

Table S.III: Studentized reports at $T = 960$

Method	Mean	SD	2.5%	Median	97.5%
Moderate-sensitivity report					
Oracle HAC	-0.028	1.008	-1.990	-0.034	1.928
Fixed metric	-0.034	1.209	-2.399	-0.040	2.322
No cross term	-0.029	1.012	-2.009	-0.034	1.971
Complete HAC	-0.029	1.018	-2.007	-0.034	1.962
High-sensitivity report					
Oracle HAC	-0.030	0.973	-1.971	-0.037	1.874
Fixed metric	-0.036	1.132	-2.259	-0.044	2.181
No cross term	-0.032	0.988	-1.984	-0.038	1.880
Complete HAC	-0.031	0.993	-1.993	-0.037	1.899

Notes: Entries summarize $(\hat{\phi} - \phi_0)/\widehat{\text{se}}$ over 2,000 replications. Standard-normal benchmarks are mean zero, SD one, and 2.5% and 97.5% quantiles -1.960 and 1.960 . The accompanying CSV retains the additional quantiles.

Table S.IV: Long-run variance channels

Report	T	Return	Metric	Cross	Complete
Moderate	240	0.00557	0.00227	0.00002	0.00786
High	240	0.01029	0.00303	-0.00009	0.01323
Moderate	960	0.00568	0.00232	0.00000	0.00800
High	960	0.01038	0.00312	-0.00001	0.01348

Notes: Entries are averages of feasible Bartlett HAC estimates. Cross denotes twice the signed return–metric long-run covariance. Complete equals Return plus Metric plus Cross, up to displayed rounding. The complete date influence is used for inference.

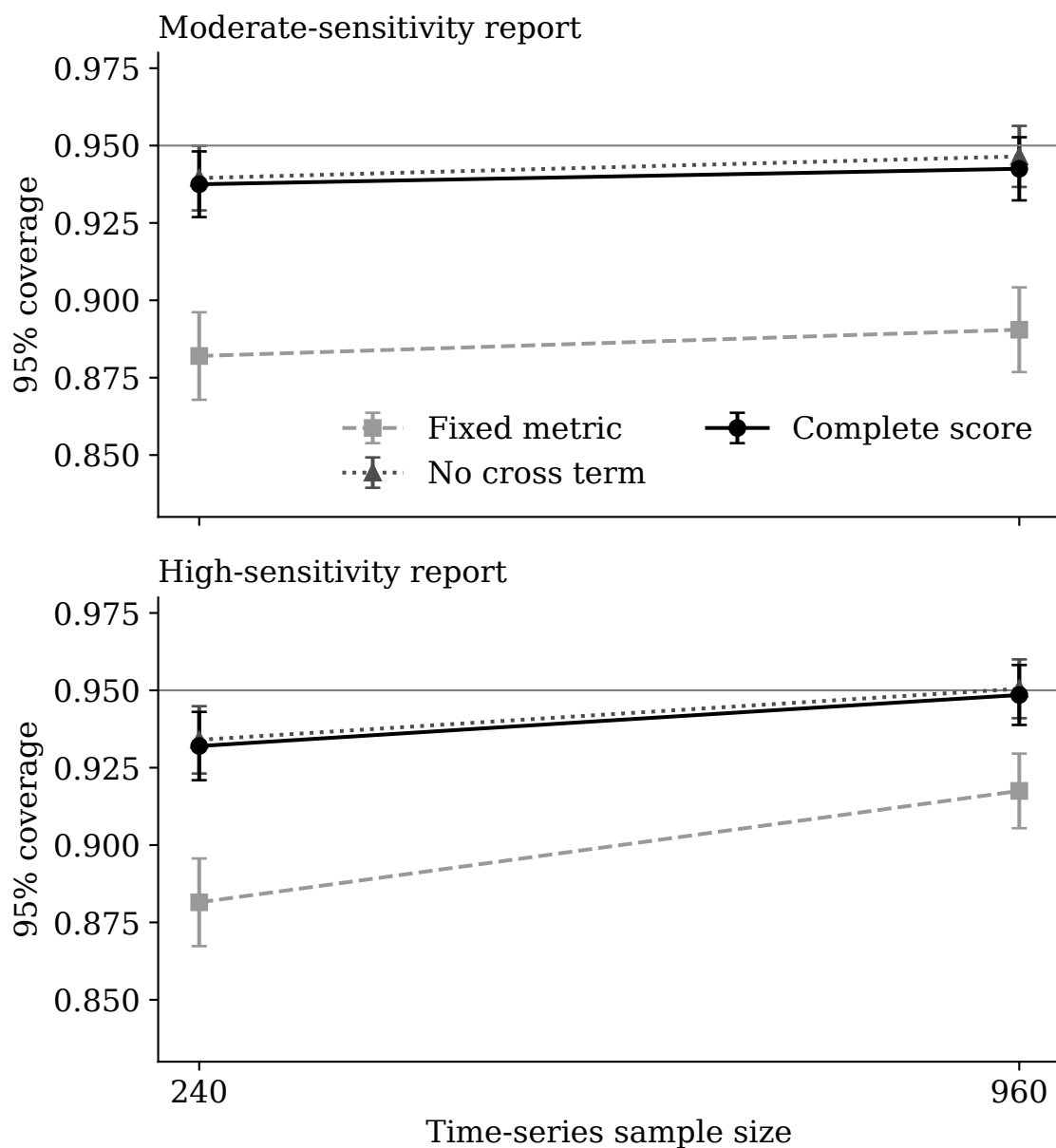


Figure S.1: Coverage for the two frozen projector reports. Each horizon uses 2,000 replications and one point estimator across covariance methods. Bars are pointwise 95 percent Monte Carlo intervals for estimated coverage; they quantify simulation error, not uncertainty in a single report.

F Learned-Boundary Simulation Details

F.1 A local experiment, not a raw panel

The simulation generates date contributions $y_t \in \mathbb{R}^3$ from a stationary Gaussian AR(1) with persistence 0.35 and marginal covariance

$$\Sigma_y = \begin{pmatrix} 1 & 0.2 & 0.5 \\ 0.2 & 0.85 & -0.3 \\ 0.5 & -0.3 & 1.1 \end{pmatrix}.$$

The first two entries are the complete score, and the last is the metric influence. The innovation covariance is $(1 - 0.35^2)\Sigma_y$; the long-run covariance is $(1 + 0.35)\Sigma_y/(1 - 0.35)$. The experiment fixes

$$H = \begin{pmatrix} 1.4 & 0.25 \\ 0.25 & 0.9 \end{pmatrix}, \quad \gamma_\eta = \begin{pmatrix} 1 \\ -0.4 \end{pmatrix}, \quad \gamma_Q = 1.5, \quad (F.1)$$

$$\phi_\eta = (0.7, -0.3), \quad \phi_Q = 0.45.$$

These choices preserve strong curvature while making the learned boundary and direct metric-report channels nontrivial.

The constrained local estimate uses the true derivatives and the realized normalized sums of y_t . For feasible calibration, a separate nine-dimensional stationary AR(1) auxiliary process with persistence 0.20 and identity marginal covariance perturbs the derivative estimates through its sample means. The scales are 0.15 for the Hessian and loading normal, 0.10 for the metric normal and loading-report derivative, and 0.08 for the direct metric-report derivative. Thus these estimation errors vanish at the root- T rate.

The numerical Hessian floor is 0.20, below the smallest eigenvalue of the true H , so it is asymptotically inactive in this design. The archived covariance routine clips eigenvalues at 10^{-10} . Here the limiting joint covariance is strictly positive definite, so this floor is also asymptotically inactive; it is not a general prescription for singular covariance limits.

F.2 Sampling and comparison rules

At each $T \in \{240, 960\}$ there are 2,000 Monte Carlo samples and 1,499 conditional Gaussian draws per sample. The seed is 2026082301. The evaluation drifts are $-4, -2, -1, -0.5, 0$.

Common draws and estimated derivatives are used for all drift comparisons within a sample. The archived envelope uses 81 points on $[-8, 0]$ and the interior endpoint. For the scalar map, this equals the two-endpoint envelope proved in the main paper.

Interior Gaussian inference ignores the projection. Fixed-boundary inference knows the true drift but sets $\gamma_Q = 0$ when calibrating the projection. Exact-boundary calibration uses drift zero throughout. Oracle local-drift inference knows the true drift and retains the learned boundary. Drift-robust inference envelopes the learned-boundary quantiles. The point estimator is held fixed across these methods; only its reference distribution changes.

The shorter-sample comparison complements the main paper's $T = 960$ exhibit. Its coverage and length panels should be read together: intervals that omit the boundary or drift uncertainty can be short because they under-cover, while the robust envelope pays for protection with length. The tables retain both nominal levels at both horizons. These are approximations to the stated local law, not a raw-panel optimizer exercise.

Table S.V: Learned-boundary coverage at $T = 240$

κ	No projection	Fixed boundary	Binding only	Known drift	Drift robust
Nominal 90% intervals					
-4	85.85	85.80	74.35	85.60	90.65
-2	88.40	87.35	78.65	86.70	93.50
-1	88.50	86.30	84.40	86.50	94.65
-0.5	86.40	84.55	86.55	86.65	94.60
0	81.65	82.25	87.00	87.00	91.80
Length at $\kappa = 0$	2.252	1.827	2.083	2.083	2.680
Nominal 95% intervals					
-4	92.00	91.90	81.40	91.70	95.20
-2	93.75	92.65	85.90	92.10	97.00
-1	93.50	91.40	90.15	92.00	97.75
-0.5	92.00	90.60	91.85	92.50	97.40
0	88.15	88.60	92.90	92.90	96.05
Length at $\kappa = 0$	2.681	2.184	2.501	2.501	3.139

Notes: Coverage is in percent, from 2,000 replications and 1,499 Gaussian draws. Length is mean interval length multiplied by $\sqrt{T}$, evaluated only at $\kappa = 0$. No projection uses a Gaussian interval; Fixed boundary knows the drift but omits the metric boundary shift; Binding only calibrates at zero; Known drift retains the learned boundary and true drift; Drift robust protects over unknown drift. All methods use one point estimator. Coverage Monte Carlo standard error near 95 percent is 0.5 percentage points.

Table S.VI: Learned-boundary coverage at $T = 960$

κ	No projection	Fixed boundary	Binding only	Known drift	Drift robust
Nominal 90% intervals					
-4	87.20	87.15	76.60	86.85	92.25
-2	89.65	88.75	80.45	87.80	94.90
-1	89.90	87.70	86.40	88.30	96.25
-0.5	88.20	86.55	88.65	88.45	95.80
0	83.20	83.90	89.80	89.80	94.40
Length at $\kappa = 0$	2.330	1.894	2.153	2.153	2.771
Nominal 95% intervals					
-4	93.20	93.20	83.80	92.90	96.30
-2	94.95	93.85	87.35	93.30	98.20
-1	94.90	92.95	92.20	93.40	98.75
-0.5	93.15	91.80	94.30	94.35	98.45
0	90.15	91.20	94.70	94.70	97.10
Length at $\kappa = 0$	2.775	2.261	2.584	2.584	3.241

Notes: Coverage is in percent, from 2,000 replications and 1,499 Gaussian draws. Length is mean interval length multiplied by $\sqrt{T}$, evaluated only at $\kappa = 0$. No projection uses a Gaussian interval; Fixed boundary knows the drift but omits the metric boundary shift; Binding only calibrates at zero; Known drift retains the learned boundary and true drift; Drift robust protects over unknown drift. All methods use one point estimator. Coverage Monte Carlo standard error near 95 percent is 0.5 percentage points.

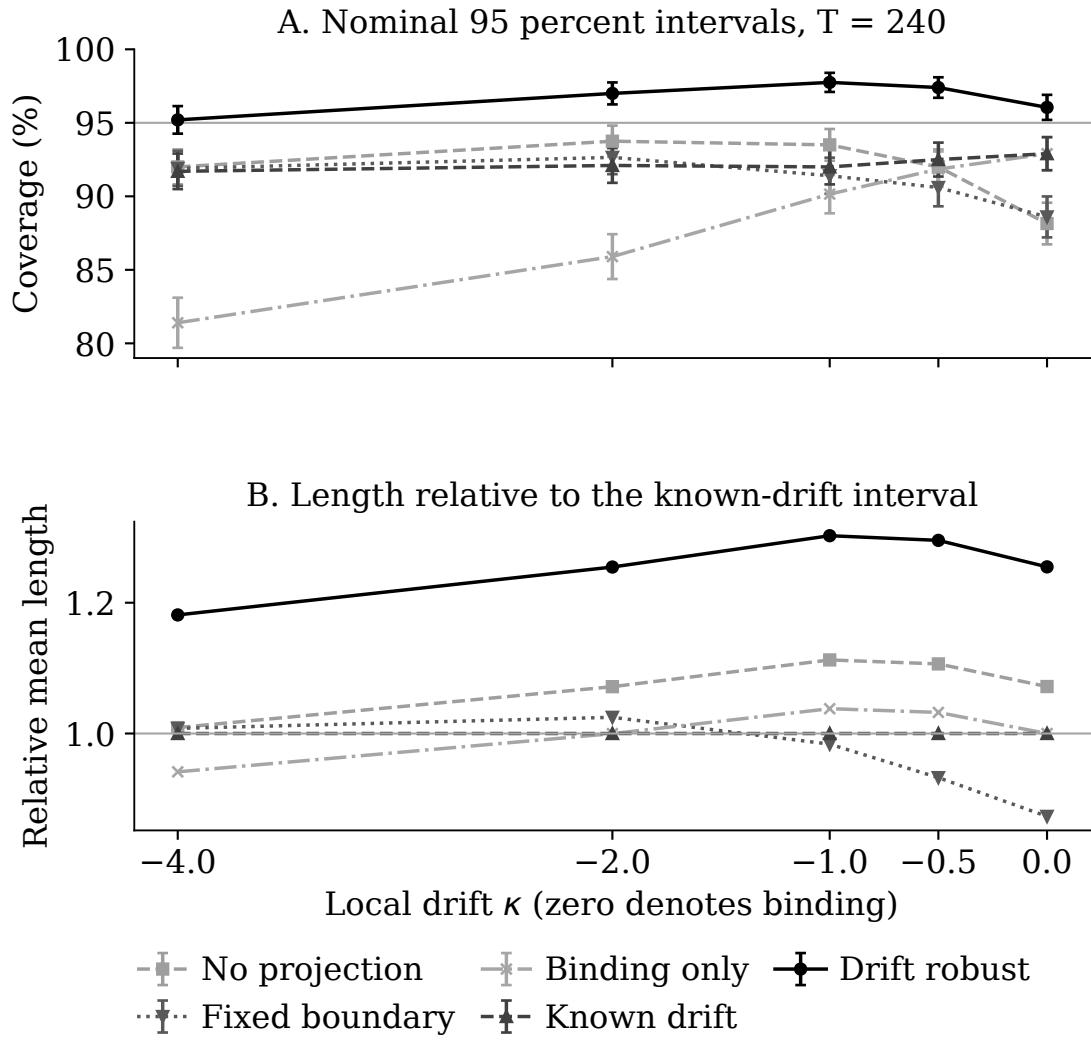


Figure S.2: Coverage and relative length at $T = 240$. Panel A reports nominal 95 percent coverage, with pointwise 95 percent Monte Carlo error bars. Panel B divides mean interval length by the known-drift mean length at the same drift. Each point uses 2,000 replications and 1,499 Gaussian draws; lines join the evaluated drifts.

G Implementation and Reproduction Map

The replication package separates three evidence sources. The calibration directory rebuilds geometric diagnostics from the retained 153-dimensional Gram matrix. The interior directory contains the frozen synthetic design, exact profile and quotient code, numerical initializer, polishing routine, coverage runner, and archived replication records. The boundary directory contains the complete local-experiment generator, projection simulation, and archived outputs.

The theory stacks retained loadings as $[\Gamma_\delta, \Gamma_\alpha]$, whereas the interior implementation stacks $[\Gamma_\alpha, \Gamma_\delta]$ with its regressor vector ordered accordingly. This fixed permutation changes neither the candidate index nor any reported statistic. It must nevertheless be respected when passing derivatives between notation and code.

For each fitted sample the inferential sequence is:

1. Select and solve the exactly profiled loading problem, checking feasibility, score accuracy, and local curvature.
2. Evaluate report and metric derivatives in one sample reference chart.
3. Form the complete date influences before estimating long-run covariance.
4. Use an interior nondegenerate scalar interval or, when justified by the maintained boundary experiment, a joint Gaussian projection interval.

A sample reference chart can put its own estimate at zero. That zero is a coordinate convention, not zero estimation uncertainty. The derivative and influence calculations quantify uncertainty in the represented class and report.

The package records input hashes, software versions, and deterministic seeds, and provides separate commands for archived-output reconstruction, unit checks, fresh small runs, and full simulation. An archived record is not a new run, and a derivative test is not a coverage experiment. The validation report distinguishes these checks.

The empirical calibration matrix is a derived input. Reconstructing it from original stock-characteristic observations requires the original panel, preprocessing information, and the applicable access permissions. The data availability statement identifies this limitation separately from the self-generated simulation data. The bundled toolkit retains its existing research license; external redistribution must follow the accompanying rights statement.

References

- Donald W. K. Andrews. Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2):399–405, 2000.
- Zheng Fang and Andres Santos. Inference on directionally differentiable functions. *The Review of Economic Studies*, 86(1):377–412, 2019.